

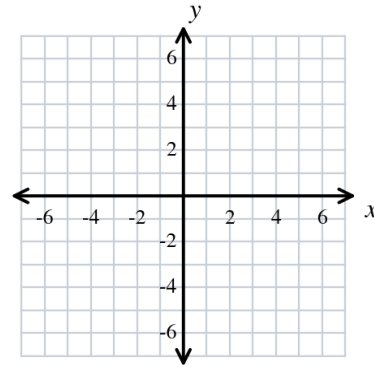
This practice is built exactly like the real test – same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. For the quadratic $y = x^2 + 10x + 9$, find the axis of symmetry, the vertex, and the y-intercept.

2. Solve by factoring: $x^2 - 5x - 14 = 0$

3. Solve by factoring: $3x^2 + 5x - 2 = 0$

4. Graph the function. Label the vertex and the axis of symmetry: $y = -x^2 + 4x - 3$



5. Solve using square roots: $5x^2 - 245 = 0$

Algebra 1 – Module 11 Practice Test

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6. Solve by completing the square: $x^2 - 10x - 11 = 0$

7. Solve using the quadratic formula: $3x^2 - 7x - 6 = 0$

8. Find the discriminant, then state how many real solutions the equation has: $2x^2 - 3x + 5 = 0$

12. A model rocket is launched from a 5-foot platform, so its height in feet after t seconds is $h = -16t^2 + 80t + 5$. Find the time when the rocket reaches its maximum height, and find that maximum height.

9. Find the zeros and the axis of symmetry: $y = (x - 3)(x + 9)$

10. Error Analysis — Gus solved $x^2 = 144$ and wrote $x = 12$. Identify his mistake, then give the full solution.

11. Solve using ANY method you choose: $x^2 - 15x + 56 = 0$

13. A rectangular garden is 7 feet longer than it is wide. Its area is 60 square feet. Write and solve a quadratic equation to find the garden's dimensions.

Cumulative Review — ACT Style. Circle the best answer.

14. Which is the COMPLETE factorization of $4x^2 + 12x + 8$?

- (A) $(4x + 4)(x + 2)$
- (B) $2(2x + 2)(x + 2)$
- (C) $4(x + 1)(x + 2)$
- (D) $(x + 2)(x + 4)$

15. The population of a colony is modeled by $y = 500(1.45)^x$. What is the percent INCREASE each year?

- (E) 145%
- (F) 45%
- (G) 1.45%
- (H) 55%

16. Simplify: $10/\sqrt{5}$

- (A) $10\sqrt{5}$
- (B) $\sqrt{5}/10$
- (C) $2\sqrt{5}$
- (D) $5\sqrt{5}$